G H Raisoni International Skill Tech University, Pune

F.Y. B. Tech (TERM-I)

CAE II- WINTER 2025 (2024 Pattern)

Indian Knowledge System (23UHSL1202)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	Find the adjoint of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}$	[5]	3	3
b)	Factorize the polynomials a) $2x^3 - 3x^2 + x - 10 = 0$ b) $x^3 - 2x^2 - 23x + 60 = 0$	[5]	3	2
	OR			
Q 2.(a)	Factorize the polynomials a) $x^3 + x^2 + x + 1 = 0$ b) $x^3 - 4x^2 + x - 8 = 0$	[5]	3	2
(b)	Find the inverse of the matrix $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$	[5]	3	3
Q 3.(a)	Find Square of the following numbers a) 113 b) 121 c) 83	[5]	4	2
(b)	Find Square root of the following numbers a) 7569 b) 7056 c) 2916	[5]	4	3
	OR			
Q. 4 (a)	Find cube of the following numbers a) 34 b) 42 c) 12 using the Vedic approach with each step	[5]	4	2
(b)	Find cube root of the following numbers a) 32768 b) 571787 c) 175616	[5]	4	3